

AI Document Translator Pro

A Comprehensive Solution for Multilingual OCR and Translation

Project Overview

AI Document Translator Pro is a professional-grade software tool developed as a B.Tech project. It integrates Optical Character Recognition (OCR), Artificial Intelligence for translation, and document processing engines to provide a seamless workflow for translating images, PDFs, Word documents, and text files into over 15 languages, including Hindi and Gujarati.

Key Features

- **Advanced OCR:** Utilizes Tesseract 5.0 to extract text from images (PNG, JPG, JPEG) with support for Indic scripts.
- **Neural Translation:** Powered by Google Neural Machine Translation (NMT) for context-aware and natural-sounding translations.
- **Batch File Support:** Ability to upload and process multiple PDF, Word, and Text files simultaneously without content mixing.
- **Smart Chunking:** Automatically splits large documents into manageable segments to bypass API limits and ensure stability.
- **Indic Script Rendering:** Built-in support for rendering Gujarati and Hindi in PDFs using embedded Lohit TrueType fonts.
- **Real-time Analytics:** Live word count tracking for both source and translated content.
- **Responsive UI:** A modern, multi-threaded Indigo & Slate themed interface that prevents application freezing during processing.

Technical Stack

- **GUI Framework:** Tkinter (Python)
- **OCR Engine:** Tesseract 5.0 (PyTesseract)
- **Translation:** Google Translate API (Deep-Translator)
- **Language Detection:** Langdetect
- **Document Processing:** Python-Docx, Pdfplumber
- **PDF Generation:** ReportLab (with TTF Embedding)

Installation & Setup

1. Install Python 3.x on your system.
2. Install required libraries: `pip install pytesseract python-docx pdfplumber deep_translator reportlab langdetect Pillow`
3. Download and install **Tesseract-OCR** and update the path in `AI.py`.
4. Place `Lohit-Gujarati.ttf` and `Lohit-Devanagari.ttf` in the project directory.
5. Run the application using: `python AI.py`

Developed by **Om Bhatt**

AI Document Translator Pro - Final Semester Project